

Exercise 1: Control Structures

TEJASWI.T

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Scenario 1: Apply Discount to Senior Citizens

Question:

Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

Creation:

```
1 CREATE TABLE Customers1 (  
2     CustomerID NUMBER PRIMARY KEY,  
3     CustomerName VARCHAR2(50),  
4     Age NUMBER,  
5     Balance NUMBER,  
6     IsVIP VARCHAR2(10)  
7 );  
8 CREATE TABLE Loans1 (  
9     LoanID NUMBER PRIMARY KEY,  
10    CustomerID NUMBER,  
11    InterestRate NUMBER,  
12    DueDate DATE  
13 );  
14
```

Insertion:

```
INSERT INTO Customers1 VALUES (201, 'Tejaswi', 68, 18000, 'FALSE');  
INSERT INTO Customers1 VALUES (202, 'Ananya', 42, 9000, 'FALSE');  
INSERT INTO Customers1 VALUES (203, 'Keerthana', 72, 25000, 'FALSE');  
  
INSERT INTO Loans1 VALUES (6001, 201, 8, SYSDATE + 10);  
INSERT INTO Loans1 VALUES (6002, 202, 9, SYSDATE + 40);  
INSERT INTO Loans1 VALUES (6003, 203, 7, SYSDATE + 20);  
  
COMMIT;
```

PL/SQL Code:

```
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1  DECLARE
2      CURSOR cust_cursor IS
3      SELECT CustomerID, Age
4      FROM Customers1;
5      BEGIN
6          FOR cust IN cust_cursor LOOP
7              IF cust.Age > 60 THEN
8                  UPDATE Loans1
9                  SET InterestRate = InterestRate - 1
10                 WHERE CustomerID = cust.CustomerID;
11              END IF;
12          END LOOP;
13          COMMIT;
14          END;
15          /
16          SELECT * FROM Loans1;
17
```

Output:

Query result					
Script output					
DBMS output					
Explain Plan					
SQL history					
🗑️ ⓘ Download ▼ Execution time: 0.001 seconds					
	LOANID	CUSTOMERID	INTERESTRATE	DUE DATE	
1	6001	201	1	7/1/2026, 4:57:28 P	
2	6002	202	9	7/31/2026, 4:57:28	
3	6003	203	0	7/11/2026, 4:57:28	

Scenario 2: VIP Customer Promotion

Question:

Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

PL/SQL Code:

```
DECLARE
    CURSOR cust_cursor IS
        SELECT CustomerID, Balance
        FROM Customers1;
BEGIN
    FOR cust IN cust_cursor LOOP
        IF cust.Balance > 10000 THEN
            UPDATE Customers1
            SET IsVIP = 'TRUE'
            WHERE CustomerID = cust.CustomerID;
        END IF;
    END LOOP;

    COMMIT;
END;
/
SELECT * FROM Customers1;
```

Output:

Query result						Script output	DBMS output	Explain Plan	SQL history
  Download ▾ Execution time: 0.005 seconds									
	CUSTOMERID	CUSTOMERNAME	AGE	BALANCE	ISVIP				
1	201	Tejaswi	68	18000	TRUE				
2	202	Ananya	42	9000	FALSE				
3	203	Keerthana	72	25000	TRUE				

Scenario 3: Loan Due Reminder

Question:

Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

PL/SQL Code:

```
BEGIN
  FOR loan IN (
    SELECT CustomerID, LoanID, DueDate
    FROM Loans1
    WHERE DueDate BETWEEN SYSDATE AND SYSDATE + 30
  )
  LOOP
    DBMS_OUTPUT.PUT_LINE(
      'Reminder: Customer ' || loan.CustomerID ||
      ' has Loan ' || loan.LoanID ||
      ' due on ' || TO_CHAR(loan.DueDate, 'DD-MON-YYYY')
    );
  END LOOP;
END;
```

Output:

Query result

Script output

DBMS output

Explain Plan

SQL history



Reminder: Customer 201 has Loan 6001 due on 01-JUL-2026
Reminder: Customer 203 has Loan 6003 due on 11-JUL-2026

PL/SQL procedure successfully completed.